F79SP Session 20: Poisson Processes: 2nd and 3rd Definitions

Recall that in the 1st Definition, a Poisson Process is defined as a counting process with interarrival times being i.i.d. exponential variables. Here we introduce the second definition and third definition.

Poisson Process - 2nd definition

Definition 0.1 (Second definition of PP)**.** *Consider the following setup:*

- *Let* $0 \leq S_1 \leq S_2 \leq \ldots\ldots$ *be a non-decreasing sequence of random variables (" random points ") on the non-negative half line.*
- *For* $t \geq 0$*. Let* $N(t) = max\{n \geq 0 \,|\, S_n \leq t\}$ *be a counting process.*

We say that $\{N(t)\}_{t\geq 0}$ *is a Poisson Process if it satisfies PP1, i.e.:*

1. *For any* $0 \leq t_1 < t_2$*,* $N(t_1, t_2) \sim \Pi(\lambda(t_2 - t_1))$*.*
2. *For any* $l = 2, 3, \ldots$ *and for any* $0 \leq t_1 < t_2 < \cdots < t_l$*, the random variables* $N(t_1), N(t_1, t_2), \ldots, N(t_{l-1}, t_l)$ *are mutually independent.*

We also say that λ *is the rate of the Poisson process.*

Note: The First and Second definition of PP are equivalent. They implies each other.

Poisson Process - 3rd Definition

We first introduce the little 'o' notation.

- The notion $o(h)$ is a function of h such that $o(h)/h \to 0$ as $h \to 0$, so that $o(h)$ is very small even in comparison to h.
- Examples are h^2, $3h^3$, etc.

Definition 0.2 (Third definition of PP)**.** *A* (homogeneous) Poisson process $N(\cdot)$ *with* rate *(intensity)* λ *is defined by the following two properties.*

(PP3(i)) *For any* t *and* $h > 0$ *(*h *small),*

$$\begin{aligned} &\mathbf{P}(N[t, t+h] = 0) = 1 - \lambda h + o(h) && \text{as } h \to 0, \\ &\mathbf{P}(N[t, t+h] = 1) = \lambda h + o(h) && \text{as } h \to 0, \\ &\mathbf{P}(N[t, t+h] \geq 2) = o(h) && \text{as } h \to 0; \end{aligned}$$

(PP3(ii)) *The numbers of points* $N(I_1), \ldots, N(I_k)$ *occurring in* disjoint *finite intervals* $I_1, \ldots, I_k$ *are* independent *random variables.*

The condition (PP3(i)) *essentially* requires that, in a *very small* interval of length h, we have either a *single point*—with probability λh, or *no point*—with probability $1 - \lambda h$.

Figure 1: Caption

Superposition Theorem and Splitting Theorem

Here, we first discuss how we can combine PPs into a single PP, using 2nd Definition of PP.

Theorem 0.1 (Superposition Theorem)**.** *Let $N_1(\cdot), \ldots, N_k(\cdot)$ be independent Poisson processes with rate $\lambda_1, \ldots, \lambda_k$ respectively. Let the process $N(\cdot) = N_1(\cdot) + \cdots + N_k(\cdot)$ be superposition of these processes (so, for any interval I, $N(I) = N_1(I) + \cdots + N_k(I)$. Then:*

1. *$N(\cdot)$ is a Poisson process with rate $\lambda = \sum_{i=1}^{k} \lambda_i$.*
2. *Given the occurrence of a point in the process $N(\cdot)$, it belongs to any given original process $N_i(\cdot)$ with probability $\frac{\lambda_i}{\lambda}$, independently of all else.*

Theorem 0.2 (Splitting Theorem)**.** *Let $N(\cdot)$ be a Poisson Process with rate λ. Suppose that each point of $N(\cdot)$ is assigned to a new point process $N_i(\cdot)$ with probability p_i, independently of all else ($p_1 + \cdots + p_k = 1$). Then: $N_1(\cdot), N_2(\cdot), \ldots, N_k(\cdot)$ are independent Poisson Process with rate $p_1\lambda$, $p_2\lambda$, $\cdots$, $p_k\lambda$.*